Rewarding Learning

Centre Number

Candidate Number

General Certificate of Secondary Education
2018

Digital Technology

Unit 1:

(Compulsory Core)

Digital Technology

[CODE]

SPECIMEN PAPER

TIME

1 hour.

INSTRUCTIONS TO CANDIDATES

Write your Centre Number and Candidate Number in the spaces provided at the top of this page.
Write your answers in the spaces provided in this question paper.
Answer all **fourteen** questions.

INFORMATION FOR CANDIDATES

The total mark for this paper is **90**.
Figures in brackets printed down the right-hand side of pages indicate the marks awarded to each or part question.
Quality of written communication will be assessed in question **11(b) and 12**.

For Examiner's use only	
Question Number	**Marks**
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
Total Marks	

Examiner Only	
Marks	Re-mark

1 Insert the appropriate letter in the space provided.

(a) The law designed to protect against software copying is:

A The Copyright, Designs and Software Act.
B The Copyright, Designs and Patents Act.
C The Copyright, Data and Patents Act.
D The Computer, Data and Software Act.

_____ [1]

(b) Select the statement which is true about flash memory technology.

A It has no moving parts and consumes very little power.
B It operates using read write heads and is used to store data.
C It has no moving parts and requires a large power supply to operate.
D It operates using read write heads and consumes very little power.

_____ [1]

(c) Select the statement which is true about fibre optic cable.

A It carries data in pulses of light but is prone to interference.
B It carries data as electronic signals and is not prone to interference.
C It carries data as electronic signals and is prone to interference.
D It carries data in pulses of light and is not prone to interference.

_____ [1]

(d) Select the statement which describes the role of the data controller in an organisation.

A Controls the way in which personal data is processed by the organisation.
B Is the individual who provides information to the organisation for processing.
C Controls the hardware the organisation uses to process the data.
D Is responsible for making the public know about their rights.

_____ [1]

(e) Select the statement which describes the role of the Information Commissioner.

A Must seek permission from companies to store personal data.
B Is responsible for writing the Data Protection Act.
C Collects personal data for use by other companies.
D Enforces the Data Protection Act.

_____ [1]

2 Jack's school uses a form to collect data from new students. The data is used to create information that can be used by the school. Place a tick (✓) beside **two** statements that are true about information.

STATEMENT	TICK (✓)
Information is raw facts and figures that have not been given a meaning.	
Information can be produced by processing raw data.	
Information is automatically encrypted when it is produced.	
Information is raw facts and figures that have been given a context.	

[2]

Examiner Only	
Marks	Re-mark

3 Sarah's dad is a professional photographer. He sometimes has to edit the images he has taken.

Examiner Only	
Marks	Re-mark

(a) Give **two** reasons why he should save his photographs in bitmap format.

1 _____ [1]

2 _____ [1]

(b) List **two** differences between bitmap and vector-based graphics.

1 _____ [1]

2 _____ [1]

4 Many mobile phones can now record digital video and sound.

Use some of the words given below to complete the paragraph which explains the process of recording and storing digital sound files.

analogue	**memory**	**digital**	**microphone**	**speaker**

Mobile phones have a built-in _____ to record sound.

Sound is input in _____ format and is then converted into

_____ format so it can be stored in the phone's memory. [3]

Examiner Only	
Marks	Re-mark

5 Jason's laptop has the latest version of an operating system installed.

List **two** functions Jason can expect the operating system to perform.

1 _____ [1]

2 _____ [1]

Examiner Only	
Marks	Re-mark

Examiner Only	
Marks	Re-mark

6 MM Books is an online company selling e-books. A data table is used to store information about the e-books for sale. Part of the data table is shown below.

Book-ID	Genre	Author Forename	Author Surname	Price	Book Title	Date Published
B2U113372	Children	Bill	Chance	£2.99	Daredevil Dave	03/05/2012
B2U313649	Fiction	Timothy	Walsh	£7.99	Eternal Dreams	07/09/2011
B2U415985	Comedy	David	Keyes	£4.99	Watch this Space	08/02/2012
B2U629649	Science Fiction	Sharon	Robson	£5.99	Aura of Mercury	06/11/2011
B2U766299	Mystery	Amir	Relf	£4.49	Macey Investigates	24/08/2013
B2U784623	Horror	Cate	Smythe	£6.49	Within these Walls	04/04/2011
B2U814222	Fiction	Melinda	Mayse	£5.99	The Room Upstairs	20/10/2013
B2U946121	Children	Daphne	Speirs	£1.99	Kate's Castle	30/08/2012
B2U911249	Historical	Alice	Kismet	£3.99	The Young Princes	25/01/2012

(a) How many fields are shown in this part of the data table?

_____ [1]

(b) What search criterion was used to select the two records shown below?

A search produces the following output:

B2U113372	Bill	Chance	£2.99	Daredevil Dave	03/05/2012
B2U946121	Daphne	Speirs	£1.99	Kate's Castle	30/08/2012

_____

_____

_____

_____ [2]

(c) Data validation is used by MM Books to make sure the data in the data table is reasonable.

Tick (✓) **two** boxes which show common methods of data validation.

	Tick (✓)
Range Check	
Heading Check	
Sort Check	
Report Check	
Presence Check	

[2]

Examiner Only	
Marks	Re-mark

7 A landscape gardener has produced a spreadsheet to work out the cost of materials used to put up a garden fence for a customer.

	A	B	C	D	E	F
1	**Materials**	**Unit Price**	**VAT**	**Selling Price**	**Number needed**	**Total Price**
2	Fence posts	£10.50	£2.10	£12.60	11	£138.60
3	Fence Panels	£35.00	£7.00	£42.00	10	£420.00
4	Cement - bag	£11.50	£2.30	£13.80	2	£27.60
5	Sand - bag	£7.75	£1.55	£9.30	3	£27.90
6	Gravel - bag	£8.99	£1.80	£10.79	4	£43.15
7	Stones - bag	£9.45	£1.89	£11.34	1	£11.34
8			TOTAL COST OF MATERIALS			£668.59
9	VAT rate	20%				

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(a) Identify the formatting used to present the label 'TOTAL COST OF MATERIALS'

_____ [1]

(b) Write down the formula you would expect to find in cell F8.

_____ [2]

(c) The formula in cell C2 is replicated down column C. State what is meant by the term replicate.

_____

_____ [1]

(d) Explain why the cell reference B9 is used in the cell C2, instead of using B9.

_____

_____

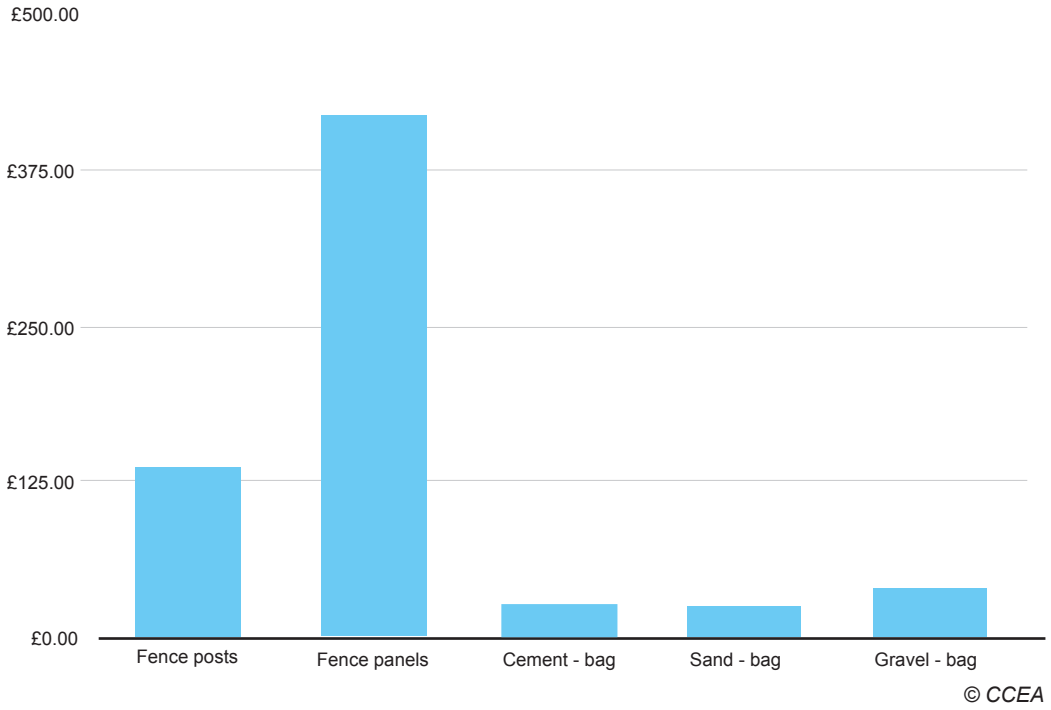
_____

_____ [2]

Examiner Only	
Marks	**Re-mark**

Examiner Only	
Marks	Re-mark

(e) Identify **two** cell ranges which have been used to create the graph shown below.



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1 _____

2 _____ [2]

(f) Explain how the gardener could use the spreadsheet model to ask 'what if' questions to help keep costs within the customer's budget.

_____

_____

_____

_____ [2]

Examiner Only	
Marks	Re-mark

8 The CPU is sometimes called the brain of the computer.

(a) Expand the acronym CPU.

_____

_____ [1]

(b) Identify the main purpose of the CPU.

_____

_____ [1]

(c) The terms given below are parts of the CPU. Complete the table by matching each part of the CPU to the statement which best matches the role it carries out.

ALU	Control Unit	Immediate Access Store

Statement	Part of the CPU that carries out this role
Holds programs and data currently in use	
Performs calculations and logical operations	
Tells input and output devices how to carry out instructions	

[3]

(d) Use the words given below to complete the paragraph to describe the role played by each of the registers when executing a program.

Memory Address Register	Memory Data Register	Program Counter

When a computer program is running the _____

holds the address of the next instruction. The location of the next

item of data to be fetched is stored in the _____ .

When a data item is fetched from memory it is stored in the

_____ . [3]

(e) Give **one** reason why ROM is suitable for storing the boot-up program.

_____

_____ [1]

(f) **(i)** List **one** possible use for cache memory.

_____

_____ [1]

(ii) Explain how the cache size can impact on CPU performance.

_____

_____

_____

_____ [2]

Examiner Only	
Marks	Re-mark

Examiner Only	
Marks	Re-mark

9 Paul uses communication technology to link electronic devices in his home to the Internet of Things (IoT).

(a) Explain what is meant by the term 'the Internet of Things'.

_____

_____

_____

_____ [2]

Paul uses Wi-Fi to link his mobile phone to the Internet when he is at home.

(b) (i) Give **one** advantage of using Wi-Fi in this way.

_____

_____ [1]

(ii) Give **two** disadvantages of using Wi-Fi in this way.

1 _____

_____ [1]

2 _____

_____ [1]

Paul has set up a LAN (Local Area Network) in his home. His devices are set up in a Bus topology.

(c) (i) Describe **two** features of a Bus topology.

1 _____

_____ [1]

2 _____

_____ [1]

(ii) Name **one** other network topology **and** describe **two** of its features.

Topology _____ [1]

Feature 1 _____

_____ [1]

Feature 2 _____

_____ [1]

Paul also uses a LAN in his job with an estate agent.

(d) List **two** advantages of using a network in an organisation.

1 _____

_____ [1]

2 _____

_____ [1]

(e) Describe the function of the following network resource.

Network Interface Card

_____

_____

_____

_____ [2]

(f) Describe what a router is and what it does.

_____

_____

_____

_____ [2]

Examiner Only	
Marks	Re-mark

10 Many computer users are worried about malware.

Examiner Only	
Marks	Re-mark

(a) In the table below tick (✓) **two** examples of malware.

Malware	Tick (✓)
Firewall	
Worm	
Router	
Virus	

[2]

Computer systems and data can be protected using encryption and firewalls.

(b) (i) Explain how encryption can be used to protect data.

_____

_____

_____

_____ [2]

(ii) List **two** ways a firewall can protect data stored on a computer.

1 _____

_____ [1]

2 _____

_____ [1]

11 Cloud computing has had an impact on the way some organisations operate.

Examiner Only	
Marks	Re-mark

(a) Explain what is meant by the term 'cloud computing'.

[2]

(b) Describe **three** advantages of cloud computing to an organisation.

Quality of written communication will be assessed in this question.

[6]

12 Cyberspace has provided a new location for crime to be committed.

Explain the term cyberspace **and** describe **two** examples of cyber crime.

Quality of written communication will be assessed in this question.

Cyberspace

Example 1

Example 2

[6]

Examiner Only	
Marks	Re-mark

13 Regular use of computers can lead to health and safety issues.

(a) List **three** measures employers must take to support safe use of computers in the workplace.

1 _____ [1]

2 _____ [1]

3 _____ [1]

The use of technology may cause health problems.

(b) Identify **two** health problems users may suffer from over use of technology.

1 _____ [1]

2 _____ [1]

Examiner Only	
Marks	Re-mark

14 Using the terms in the list below choose one term that best matches the definition in the left hand column (not all terms will be used).

Examiner Only	
Marks	Re-mark

WAN	**Intranet**	**Real time processing**	**Upskilling**	**Batch processing**
	Computer Misuse Act		**Consumer Contracts Regulations**	
GPS	**Wi-Fi**	**Bluetooth**	**Job displacement**	**Data Protection Act**

Definition	**Term**
A radio navigation system that allows users determine their exact location.	
Legislation implemented to give you rights when shopping online.	
A situation where a long-standing employee is removed from their original job and provided with an alternative job within the same organisation.	
A processing method where the input is dealt with quickly enough so as to be able to control an output properly.	
A private network or website only available to members of an organisation. Not limited by geographic location.	
A wireless technology standard for exchanging data over short distances.	

[6]